Compiler Construction

Lab 10

Name: Syeda Ayesha Fatima

Sap id: 53953

Submitted to: Ma'am Hasnat

Date: April 7th, 2026

Grammar:

S → A B
A → a A | b
B → c B | d

Task 1:

```
#include <iostream>
#include <stack>
#include <map>
#include <vector>
using namespace std;
int main() {
    map<int, map<string, string>> action;
    action[0]["a"] = "s3";
    action[0]["b"] = "s4";
    action[1]["$"] = "acc";
    action[2]["c"] = "s6";
    action[2]["d"] = "s7";
    action[3]["a"] = "s3";
    action[3]["b"] = "s4";
// LR (0) Parser
    action[4]["a"] = "r3";
    action[4]["b"] = "r3";
    action[4]["c"] = "r3";
    action[4]["d"] = "r3";
    action[4]["$"] = "r3";
    action[5]["$"] = "r1";
```

```
action[7]["a"] = "r5";
action[7]["b"] = "r5";
action[7]["c"] = "r5";
action[7]["d"] = "r5";
action[7]["$"] = "r5";
action[8]["a"] = "r2";
action[8]["b"] = "r2";
action[8]["c"] = "r2";
action[8]["d"] = "r2";
action[8]["$"] = "r2";
map<int, map<string, int>> goTo;
goTo[0]["S"] = 1;
goTo[0]["A"] = 2;
goTo[2]["B"] = 5;
goTo[3]["A"] = 8;
map<int, int> prodLen;
prodLen[1] = 2; // S → A B
prodLen[2] = 2; // A → a A
prodLen[3] = 1; // A → b
prodLen[4] = 2; // B → c B
prodLen[5] = 1; // B → d
vector<string> input;
string s;
cout << "Enter input (end with $): ";
while (cin >> s) {
    input.push_back(s);
    if (s == "$") break;
}
```

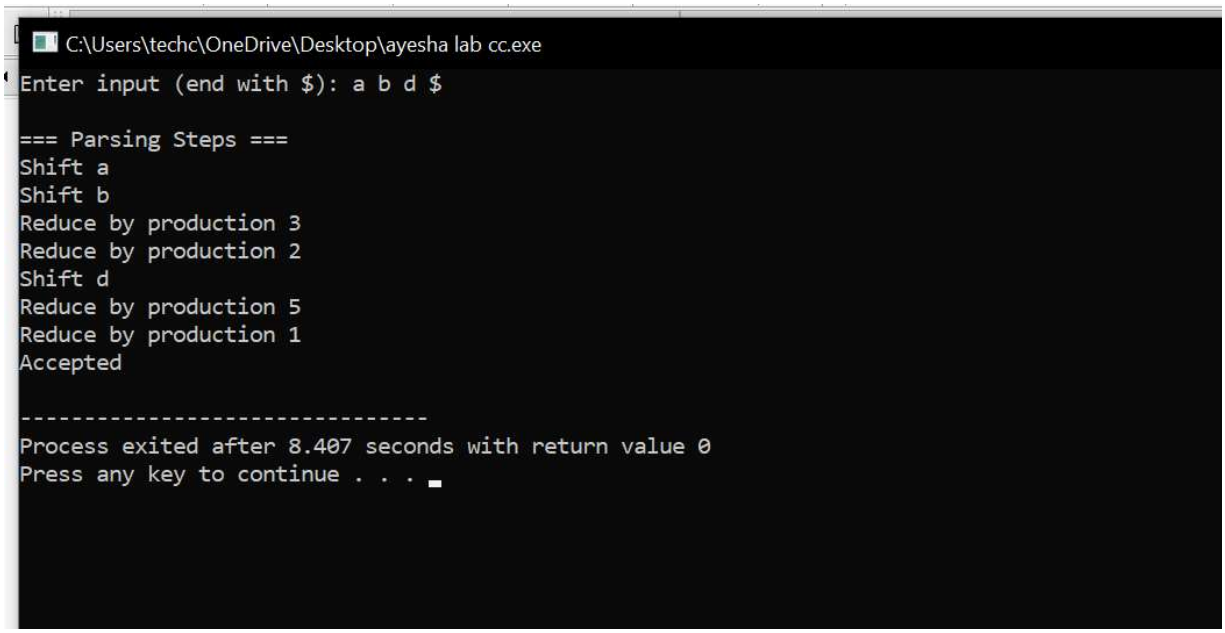
```
stack<int> st;
st.push(0);
int i = 0;

cout << "\n=== Parsing Steps ===\n";

while (true) {
    int state = st.top();
    string curr = input[i];
    string act = action[state][curr];
    if (act == "") {
        cout << "Error\n";
        break;
    }
    if (act[0] == 's') {
        int next = stoi(act.substr(1));
        st.push(next);
        cout << "Shift " << curr << endl;
        i++;
    }
    else if (act[0] == 'r') {
        int p = stoi(act.substr(1));
        int len = prodLen[p];
        for (int j = 0; j < len; j++)
            st.pop();
        int top = st.top();
        string lhs;
        if (p == 1) lhs = "S";
```

```
            else if (p == 2 || p == 3) lhs = "A";
            else lhs = "B";
            st.push(goTo[top][lhs]);
            cout << "Reduce by production " << p << endl;
        }
        else if (act == "acc") {
            cout << "Accepted\n";
            break;
        }
    }
    return 0;
}
```

Output:



Canonical LR(0) Item Sets Table

State	LR(0) Items
I0	S' → • S S → • A B A → • a A A → • b
I1	S' → S •
I2	S → A • B B → • c B B → • d
I3	A → a • A A → • a A A → • b
I4	A → b •
I5	S → A B •
I6	B → c • B B → • c B B → • d
I7	B → d •
I8	A → a A •

Transition Table

From	Symbol	To
I0	S	I1
I0	A	I2
I0	a	I3
I0	b	I4
I2	B	I5
I2	c	I6
I2	d	I7
I3	A	I8

Action Table

State	a	b	c	d	$	S	A	B
0	s3	s4				1	2	
1					acc			
2			s6	s7				5
3	s3	s4					8	
4	r3	r3	r3	r3	r3			
5					r1			
6			s6	s7				9
7	r5	r5	r5	r5	r5			
8	r2	r2	r2	r2	r2			

Task 2:

```
#include <iostream>
#include <stack>
#include <map>
#include <vector>
using namespace std;
int main() {
```

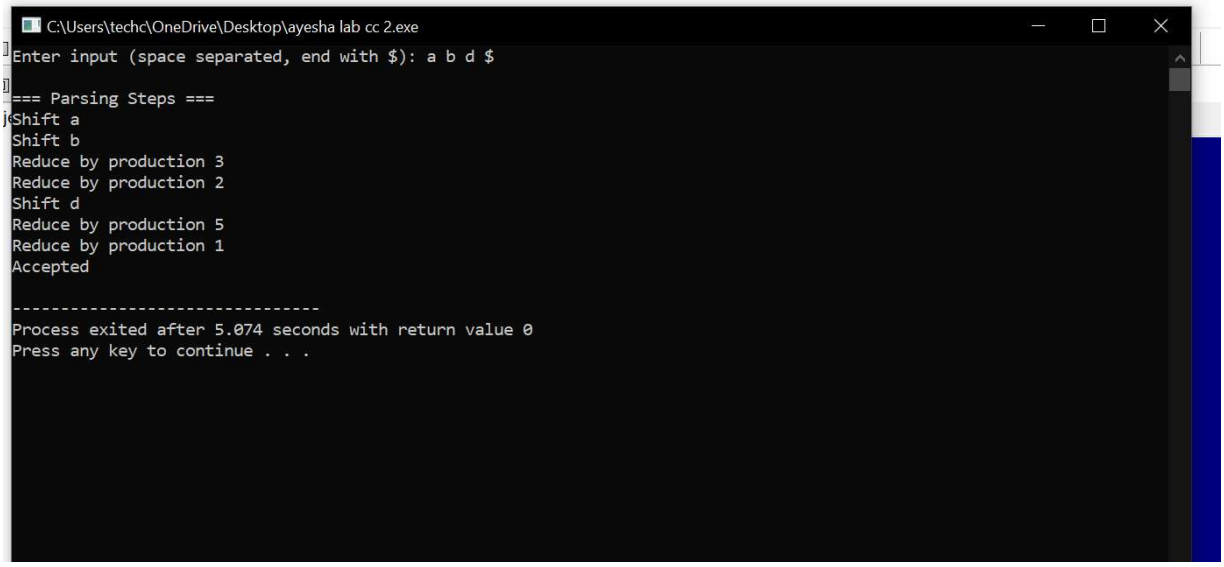
```
map<int, map<string, string>> action;
action[0]["a"] = "s3";
action[0]["b"] = "s4";
action[1]["$"] = "acc";
action[2]["c"] = "s6";
action[2]["d"] = "s7";
action[3]["a"] = "s3";
action[3]["b"] = "s4";
action[4]["c"] = "r3";
action[4]["d"] = "r3";
action[5]["$"] = "r1";
action[7]["$"] = "r5";
action[8]["c"] = "r2";
action[8]["d"] = "r2";
map<int, map<string, int>> goTo;
goTo[0]["S"] = 1;
goTo[0]["A"] = 2;
goTo[2]["B"] = 5;
goTo[3]["A"] = 8;
map<int, int> prodLen;
prodLen[1] = 2; // S → A B
prodLen[2] = 2; // A → a A
prodLen[3] = 1; // A → b
prodLen[4] = 2; // B → c B
prodLen[5] = 1; // B → d
vector<string> input;
string s;
cout << "Enter input (space separated, end with $): ";
```

```
while (cin >> s) {
    input.push_back(s);
    if (s == "$") break;
}
stack<int> st;
st.push(0);
int i = 0;
cout << "\n=== Parsing Steps ===\n";
while (true) {
    int state = st.top();
    string curr = input[i];
    string act = action[state][curr];
    if (act == "") {
        cout << "Error\n";
        break;
    }
    if (act[0] == 's') {
        int next = stoi(act.substr(1));
        st.push(next);
        cout << "Shift " << curr << endl;
        i++;
    }
    else if (act[0] == 'r') {
        int p = stoi(act.substr(1));
        int len = prodLen[p];
        for (int j = 0; j < len; j++)
            st.pop();
        int top = st.top();
```

```
            string lhs;
            if (p == 1) lhs = "S";
            else if (p == 2 || p == 3) lhs = "A";
            else lhs = "B";

            st.push(goTo[top][lhs]);
            cout << "Reduce by production " << p << endl;
        }
        else if (act == "acc") {
            cout << "Accepted\n";
            break;
        }
    }
    return 0;
}
```

Output:



Action Table

State	a	b	c	d	$	S	A	B
0	s3	s4				1	2	
1					acc			
2			s6	s7				5
3	s3	s4					8	
4			r3	r3				
5					r1			
6			s6	s7				9
7					r5			
8			r2	r2				